ETHRC Motor Controller

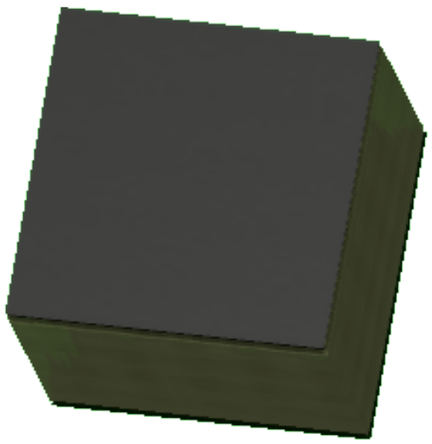
Variant: DRAFT

2026-11-08

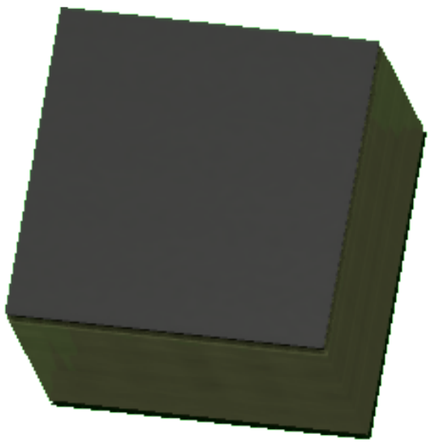
Rev 1.1.1+ (Unreleased)

Page	Index	Page	Index	Page	Index	Page	Index
1	Cover Page	11	Error: XML File not found	21	Error: XML File not found	31	Error: XML File not found
2	Error: XML File not found	12	Error: XML File not found	22	Error: XML File not found	32	Error: XML File not found
3	Error: XML File not found	13	Error: XML File not found	23	Error: XML File not found	33	Error: XML File not found
4	Error: XML File not found	14	Error: XML File not found	24	Error: XML File not found	34	Error: XML File not found
5	Error: XML File not found	15	Error: XML File not found	25	Error: XML File not found	35	Error: XML File not found
6	Error: XML File not found	16	Error: XML File not found	26	Error: XML File not found	36	Error: XML File not found
7	Error: XML File not found	17	Error: XML File not found	27	Error: XML File not found	37	Error: XML File not found
8	Error: XML File not found	18	Error: XML File not found	28	Error: XML File not found	38	Error: XML File not found
9	Error: XML File not found	19	Error: XML File not found	29	Error: XML File not found	39	Error: XML File not found
10	Error: XML File not found	20	Error: XML File not found	30	Error: XML File not found	40	Error: XML File not found

TOP VIEW



BOTTOM VIEW



NOTES

Schematic based off Josh Pieper's moteus controllers.

Not fitted components are marked as ✗

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

Date: 08-Nov-2026

DESIGN CONSIDERATIONS

DESIGN NOTE: Example text for informational design notes.

DESIGN NOTE: Example text for debug notes.

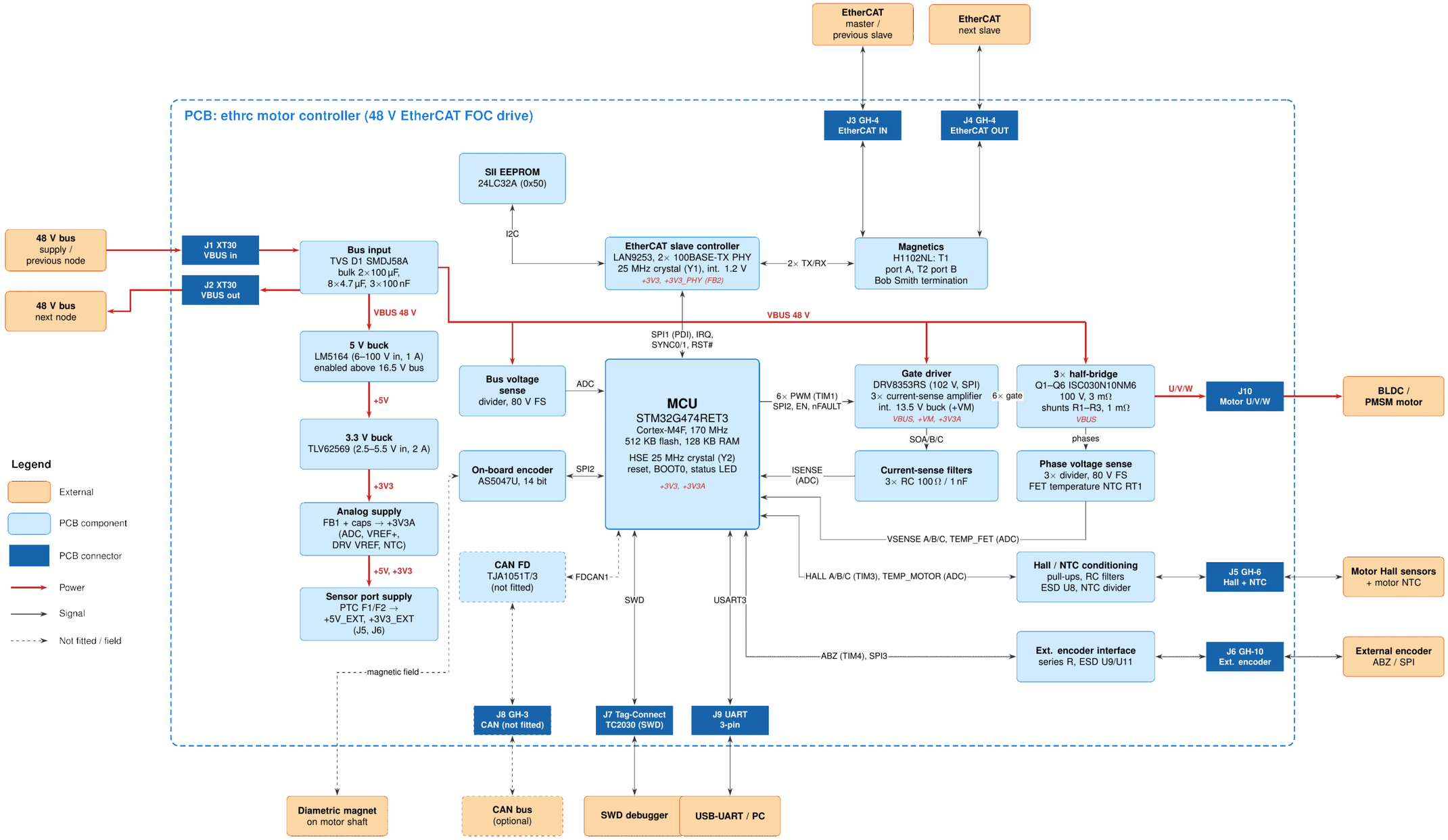
DESIGN NOTE: Example text for cautionary design notes.

DESIGN NOTE: Example text for critical design notes.

LAYOUT NOTE: Example text for critical layout guidelines.

Comments:	Company: ETH Robotics Club		Variant: DRAFT	Git Hash: 1a83c5d
	Board Name: ETHRC Motor Controller		Project Name: BLDC Motor Controller	
Sheet Title: Cover Page	File Name: ethrc_motor_controller.kicad_sch	Designer: Masiar Etemadi	Date: 2024-04-13	Revision: 1.1.1+ (Unreleased)
Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 20

[2] Block Diagram

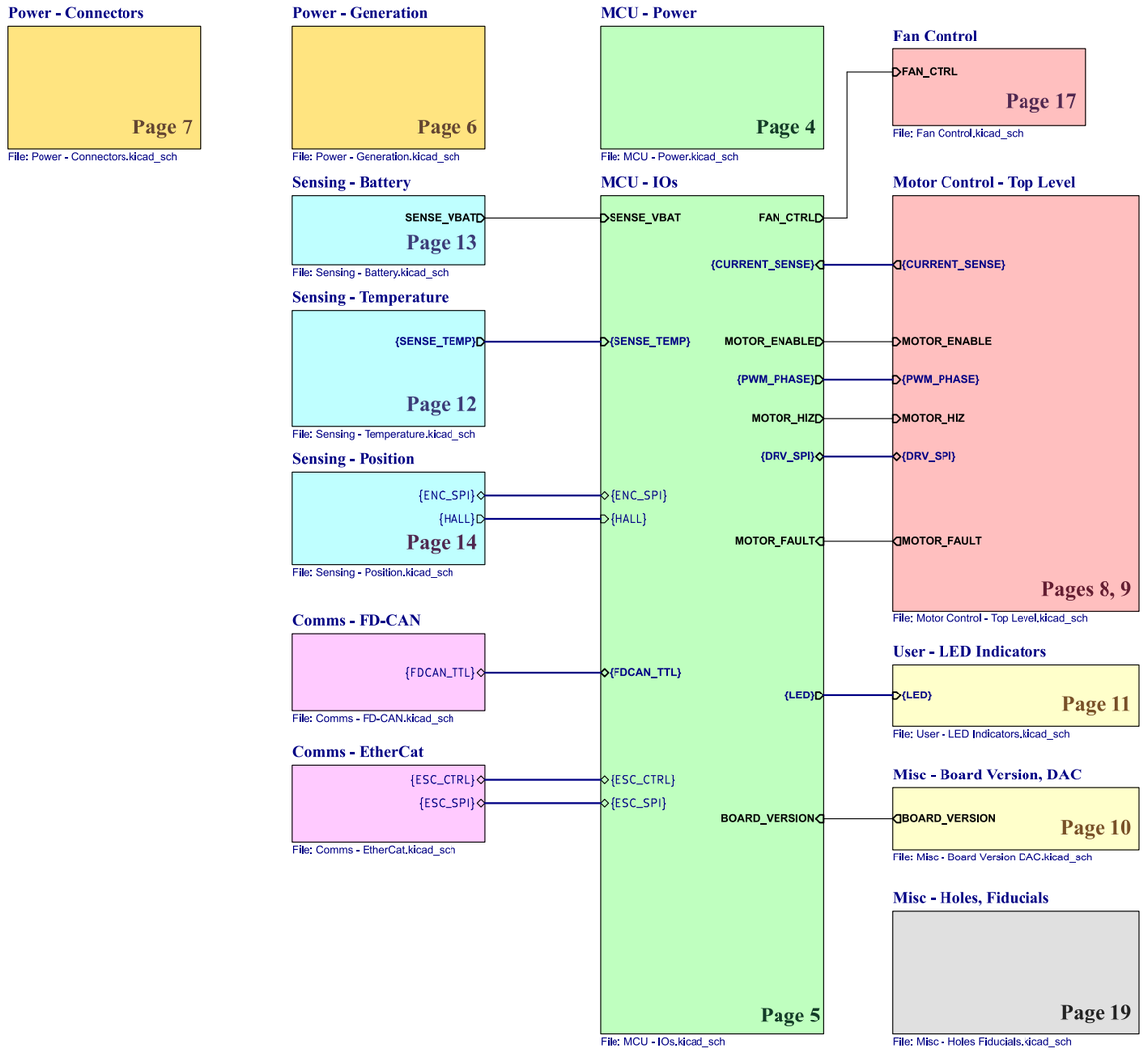


Target specifications:

Input voltage:	36-60 V
Continuous current:	25 A
Peak current:	65 A
Switching frequency:	20-60 kHz

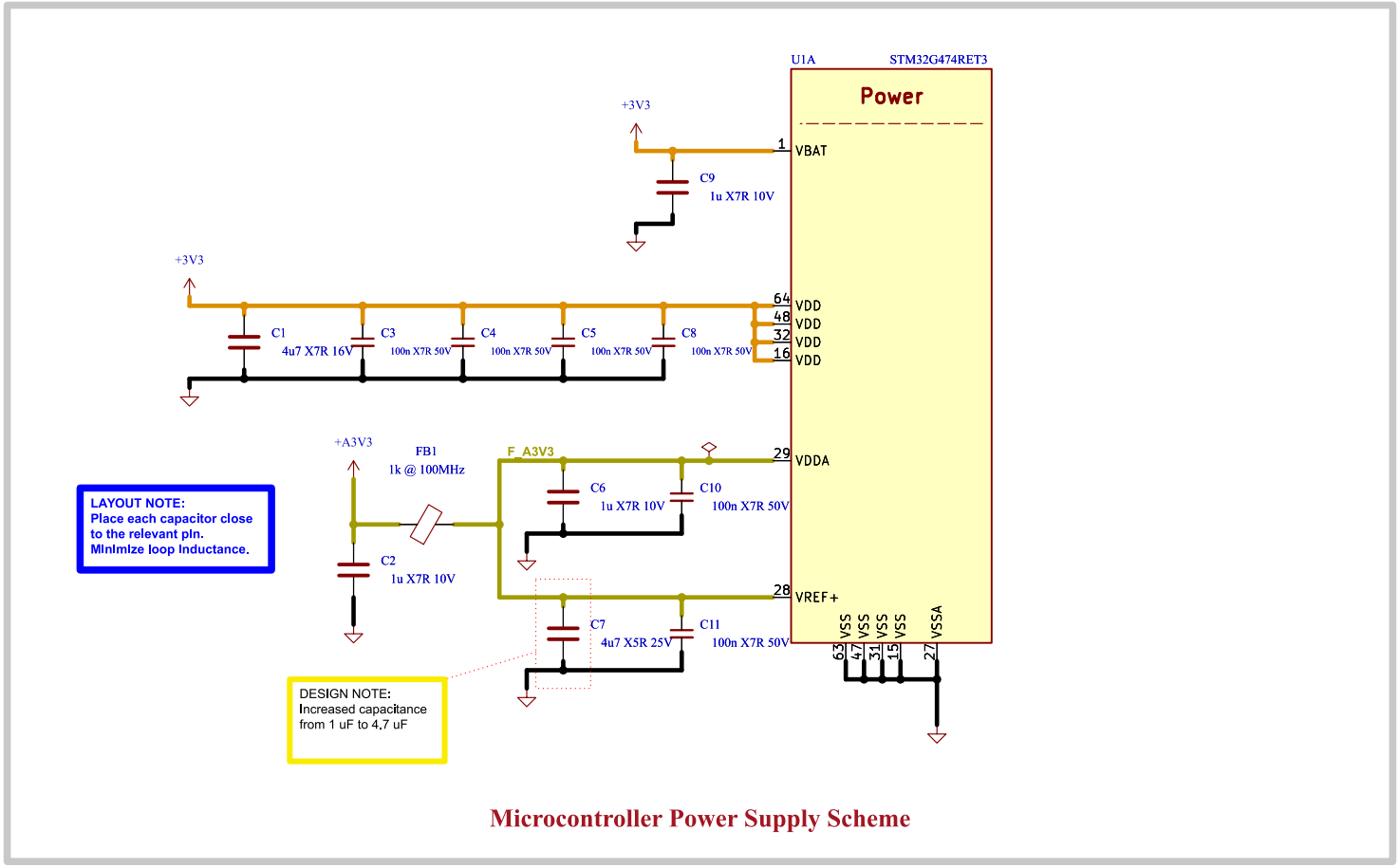
Comments:	Company: ETH Robotics Club		Variant: DRAFT	Git Hash: 1a83c5d
	Board Name: ETHRC Motor Controller		Project Name: BLDC Motor Controller	
Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Masiar Etemadi	Date: 2024-04-13	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Block Diagram/		Reviewer:	Size: A3	Sheet: 2 of 20

[3] Project Architecture



Comments:	Company: **ETH Robotics Club**		Variant: DRAFT	Git Hash: 1a83c5d
	Board Name: **ETHRC Motor Controller**		Project Name: **BLDC Motor Controller**	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Masiar Etemadi	Date: 2023-12-22	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/		Reviewer:	Size: **A3**	Sheet: **3** of **20**

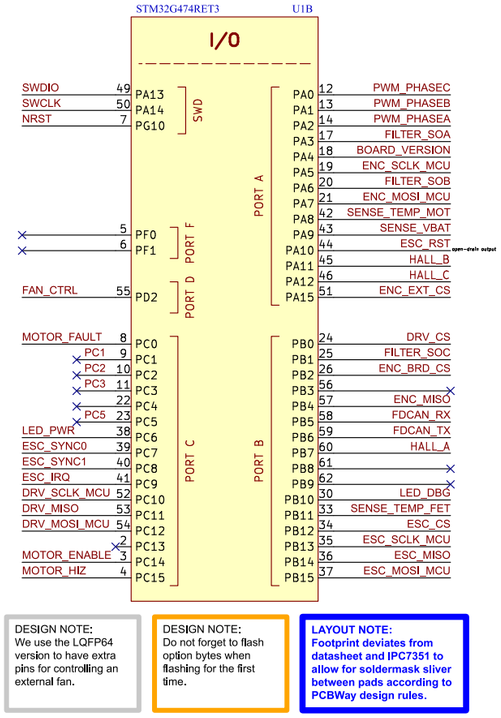
[4] MCU - Power



Microcontroller Power Supply Scheme

Comments: AN5346; STM32G474 Datasheet p.81; J. Pieper ADC investigation	Company: ETH Robotics Club		Variant: DRAFT	Git Hash: 1a83c5d
	Board Name: ETHRC Motor Controller		Project Name: BLDC Motor Controller	
Sheet Title: MCU - Power	File Name: MCU - Power.kicad_sch	Designer: Masiar Etemadi	Date: 2023-12-18	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/MCU - Power/		Reviewer:	Size: A4	Sheet: 4 of 20

[5] MCU - I/Os

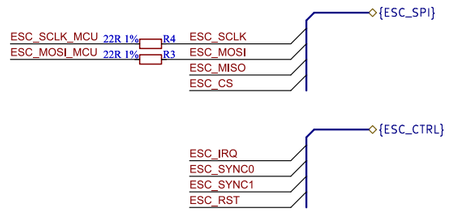


DESIGN NOTE:
We use the LQFP64 version to have extra pins for controlling an external fan.

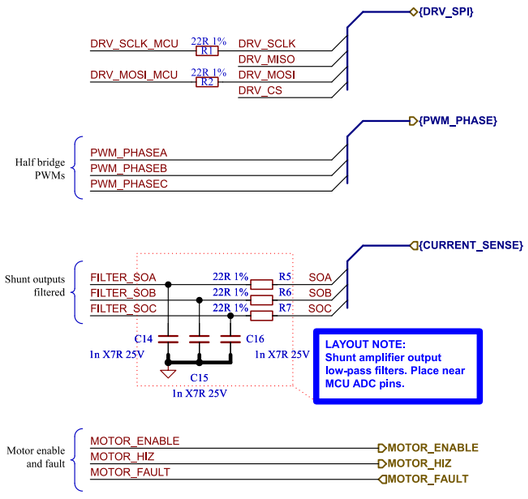
DESIGN NOTE:
Do not forget to flash option bytes when flashing for the first time.

LAYOUT NOTE:
Footprint deviates from datasheet and IPC7351 to allow for soldermask sliver between pads according to PCBWay design rules.

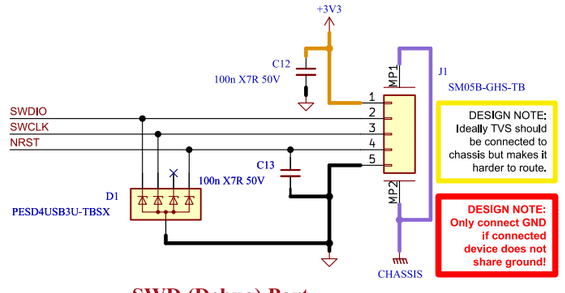
Microcontroller I/O Domain



Connection to EtherCat Subdevice Controller



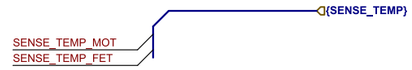
Connection to Gate Driver



SWD (Debug) Port



5Mbps FD-CAN Bus



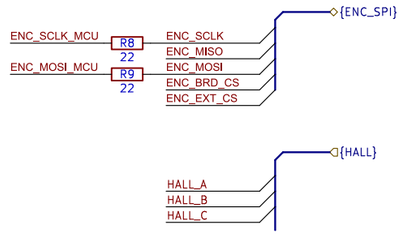
Temperature Sense



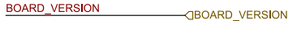
Battery Voltage Sense



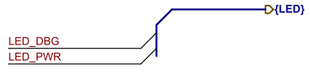
Low-Side Fan Control



Position Sense



Board Version Selection



User LEDs

Gate Driver

MOTOR_ENABLE	TP1	MOTOR_ENABLE
MOTOR_HIZ	TP2	MOTOR_HIZ
MOTOR_FAULT	TP3	MOTOR_FAULT
DRV_SCLK	TP4	DRV_SCLK
DRV_MISO	TP5	DRV_MISO
DRV_MOSI	TP6	DRV_MOSI
DRV_CS	TP7	DRV_CS
PWM_PHASEA	TP8	PWM_PHASEA
PWM_PHASEB	TP9	PWM_PHASEB
PWM_PHASEC	TP10	PWM_PHASEC
SOA	TP11	SOA
SOB	TP12	SOB
SOC	TP13	SOC

Debug

SWDIO	TP14	SWDIO
SWCLK	TP15	SWCLK
NRST	TP16	NRST

ESC_RST	TP17	ESC_RST
ESC_SCLK	TP18	ESC_SCLK
ESC_MISO	TP19	ESC_MISO
ESC_MOSI	TP20	ESC_MOSI
ESC_CS	TP21	ESC_CS
ESC_IRQ	TP22	ESC_IRQ
ESC_SYNC0	TP23	ESC_SYNC0
ESC_SYNC1	TP75	ESC_SYNC1

ENC_SCLK	TP24	ENC_SCLK
ENC_MISO	TP25	ENC_MISO
ENC_MOSI	TP26	ENC_MOSI
ENC_BRD_CS	TP27	ENC_BRD_CS
ENC_EXT_CS	TP28	ENC_EXT_CS

RS-422

LEDs

LED_DBG	TP31	LED_DBG
LED_PWR	TP32	LED_PWR

FD-CAN

FDCAN_RX	TP33	FD_CAN_RX
FDCAN_TX	TP34	FDCAN_TX

Fan

FAN_CTRL	TP35	FAN_CTRL

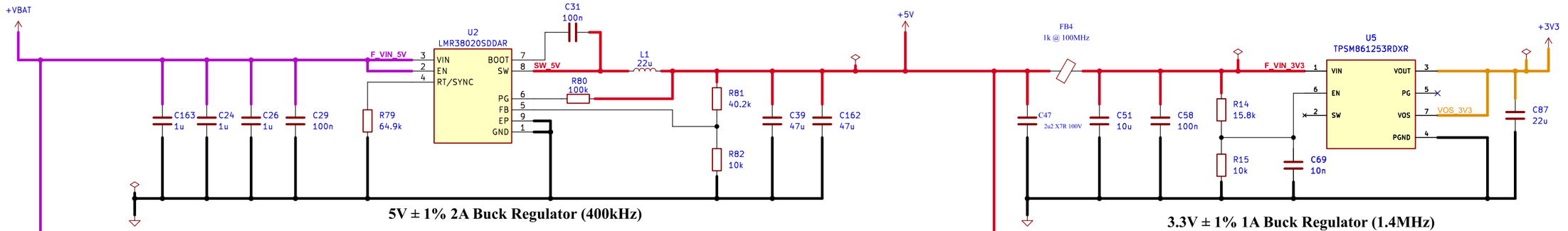
Sense

SENSE_TEMP_MOT	TP36	SENSE_TEMP_MOT
SENSE_TEMP_FET	TP37	SENSE_TEMP_FET
SENSE_VBAT	TP38	SENSE_VBAT

Test Points

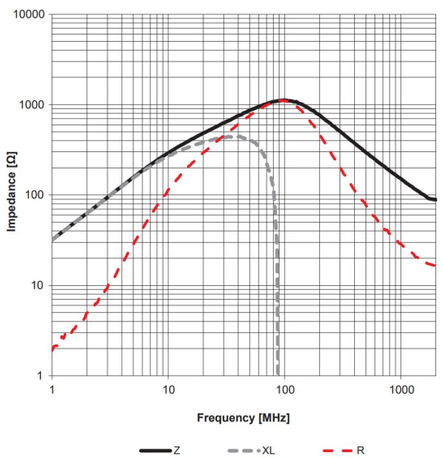
Comments: **References:** Flexible I/O worked examples Flexible I/O source configuration	Company: **ETH Robotics Club**		Variant: DRAFT	Git Hash: 1a83c5d
	Board Name: **ETHRC Motor Controller**		Project Name: **BLDC Motor Controller**	
Sheet Title: MCU - I/Os	File Name: MCU - IOs.kicad_sch	Designer: Masiar Etemadi	Date: 2023-12-20	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/MCU - IOs/		Reviewer:	Size: **A3**	Sheet: **5** of **20**

[6] Power - Generation



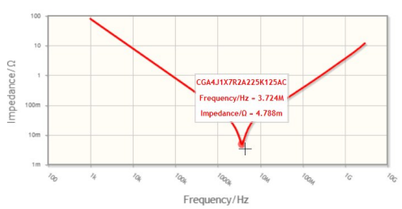
Voltage Regulators

Ferrite bead Impedance vs Frequency curve



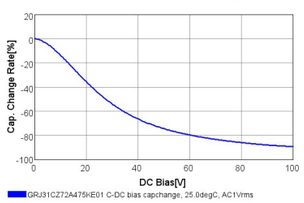
DESIGN NOTE:
XL drops around 45th harmonic.

Pi filter capacitor Impedance vs Frequency curve



CGA4J1X7R2A225K125AC

Bulk capacitance Derating curve



DESIGN NOTE:
Effective capacitance at 33V is approximately 2.1 uF

Capacitor and Ferrite Bead Parameters

Battery

+VBAT	TP39	+VBAT
+VBAT	TP40	+VBAT
+VBAT	TP41	+VBAT

Filtered Power Rails

F_VIN_12V	TP42	F_VIN_12V
F_VIN_5V	TP43	F_VIN_5V
F_VIN_3V3	TP44	F_VIN_3V3

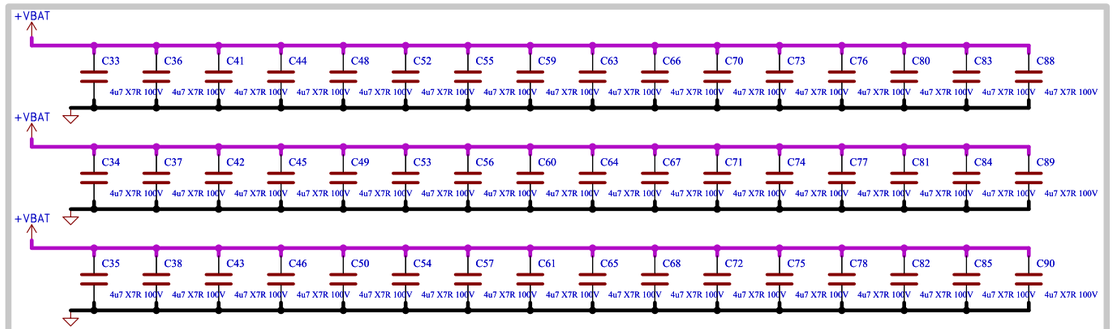
Power Rails

+12V	TP45	+12V
+5V	TP46	+5V
+5V	TP47	+5V
+3V3	TP48	+3V3
+3V3	TP49	+3V3
+3V3	TP50	+3V3
+A3V3	TP51	+A3V3

Ground

GND	TP52	GND
GND	TP53	GND
GND	TP54	GND
GND	TP55	GND
GND	TP56	GND
GND	TP57	GND
GND	TP58	GND

Test Points



DESIGN NOTE:
An external precharge circuit is advised to charge the bulk capacitors if multiple boards are connected.

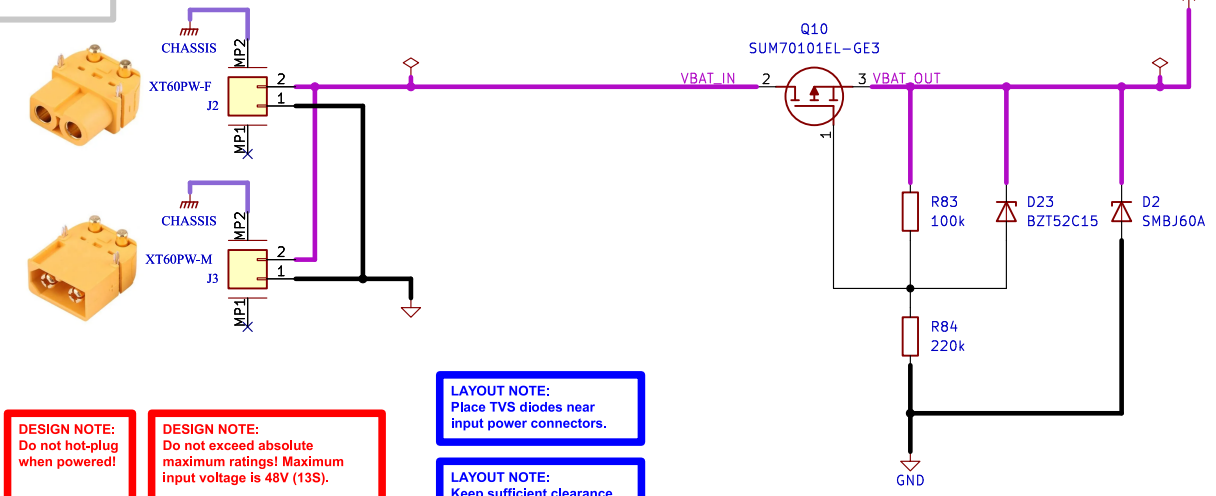
DC-Link Bulk Capacitors. 48 x 4.7 uF = 225.6 uF (100.8 uF at 33V)

DESIGN NOTE:
X6S 10uF is marginally better (but worse DC bias derating) at a higher price.

Comments: (+12V (600 mA): Fan power) +5V (600 mA): Gate Driver, LEDs +3V3 (1 A): digital 3.3V +3V3A (250 mA): analog 3.3V	Company: ETH Robotics Club	Variant: DRAFT	Git Hash: 1a83c5d
	Board Name: ETHRC Motor Controller		Project Name: BLDC Motor Controller
Sheet Title: Power - Generation	File Name: Power - Generation.kicad_sch	Designer: Masiar Etemadi	Date: 2024-04-13 / Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/Power - Generation/		Reviewer:	Size: A3 / Sheet: 6 of 20

[7] Power - Connectors

DESIGN NOTE:
It's best to power the controller through the male connector to always have a female connector at the end of the chain.



DESIGN NOTE:
Do not hot-plug when powered!

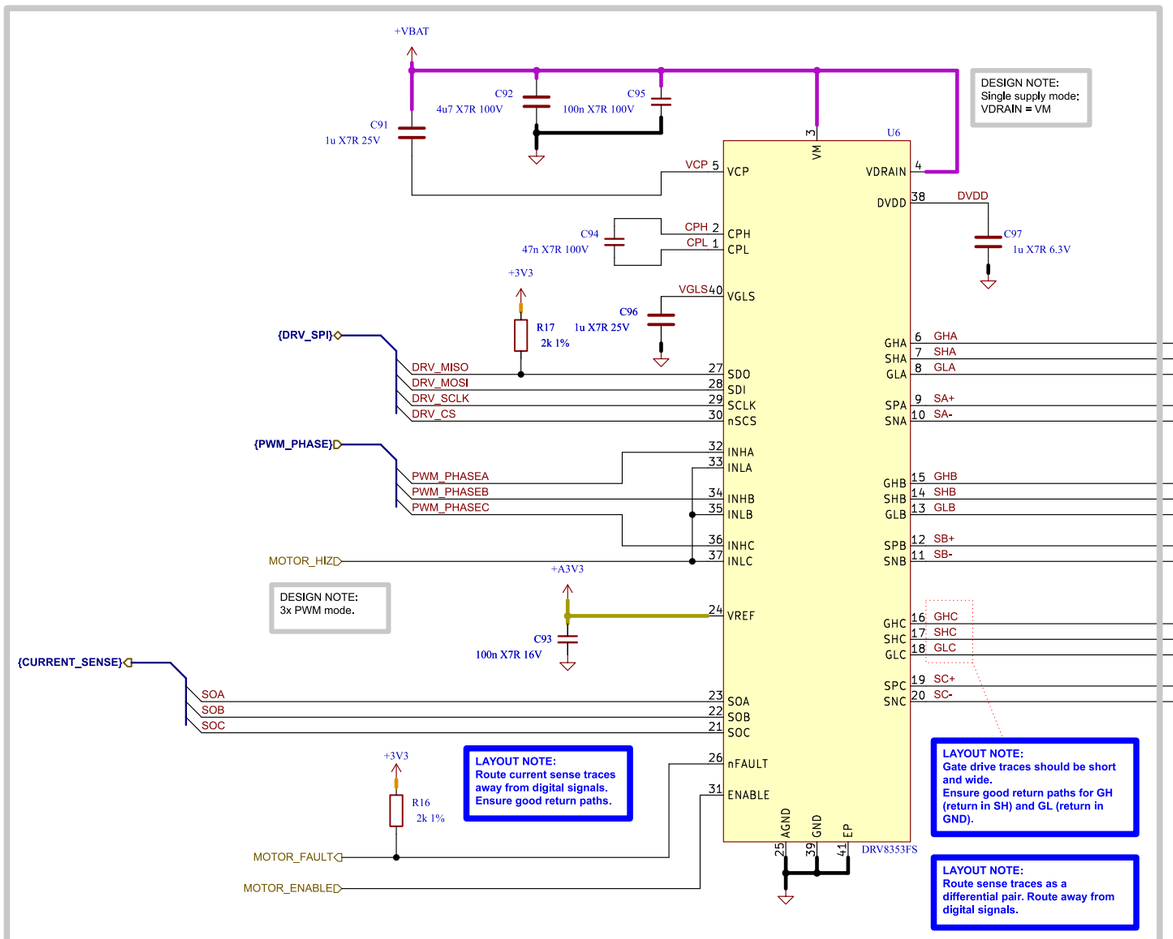
DESIGN NOTE:
Do not exceed absolute maximum ratings! Maximum input voltage is 48V (13S).

LAYOUT NOTE:
Place TVS diodes near input power connectors.

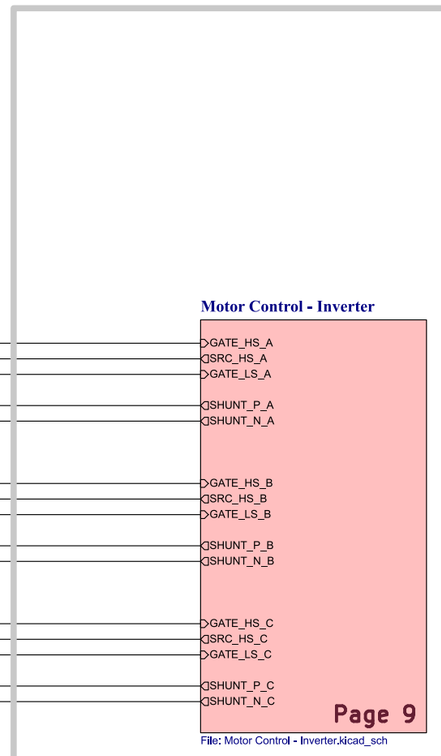
LAYOUT NOTE:
Keep sufficient clearance between power nets according to IPC-2221/IEC60664-1

Comments:	Company: ETH Robotics Club		Variant: DRAFT	Git Hash: 1a83c5d
	Board Name: ETHRC Motor Controller		Project Name: BLDC Motor Controller	
Sheet Title: Power - Connectors	File Name: Power - Connectors.kicad_sch	Designer: Masiar Etemadi	Date: 2023-12-31	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/Power - Connectors/		Reviewer:	Size: A4	Sheet: 7 of 20

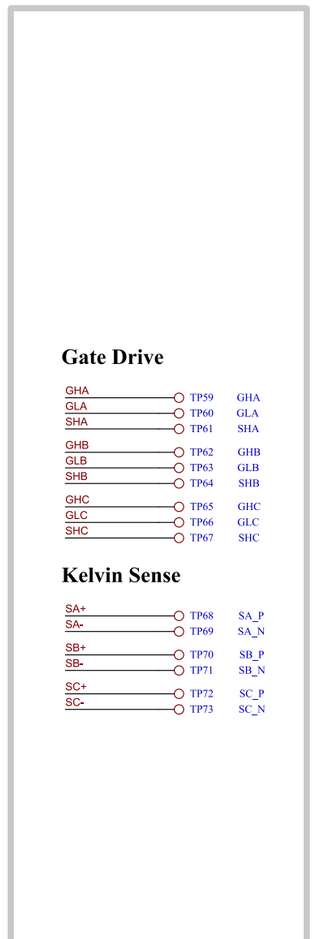
[8] Motor Control - Top Level



100-V Three-Phase Smart Gate Driver



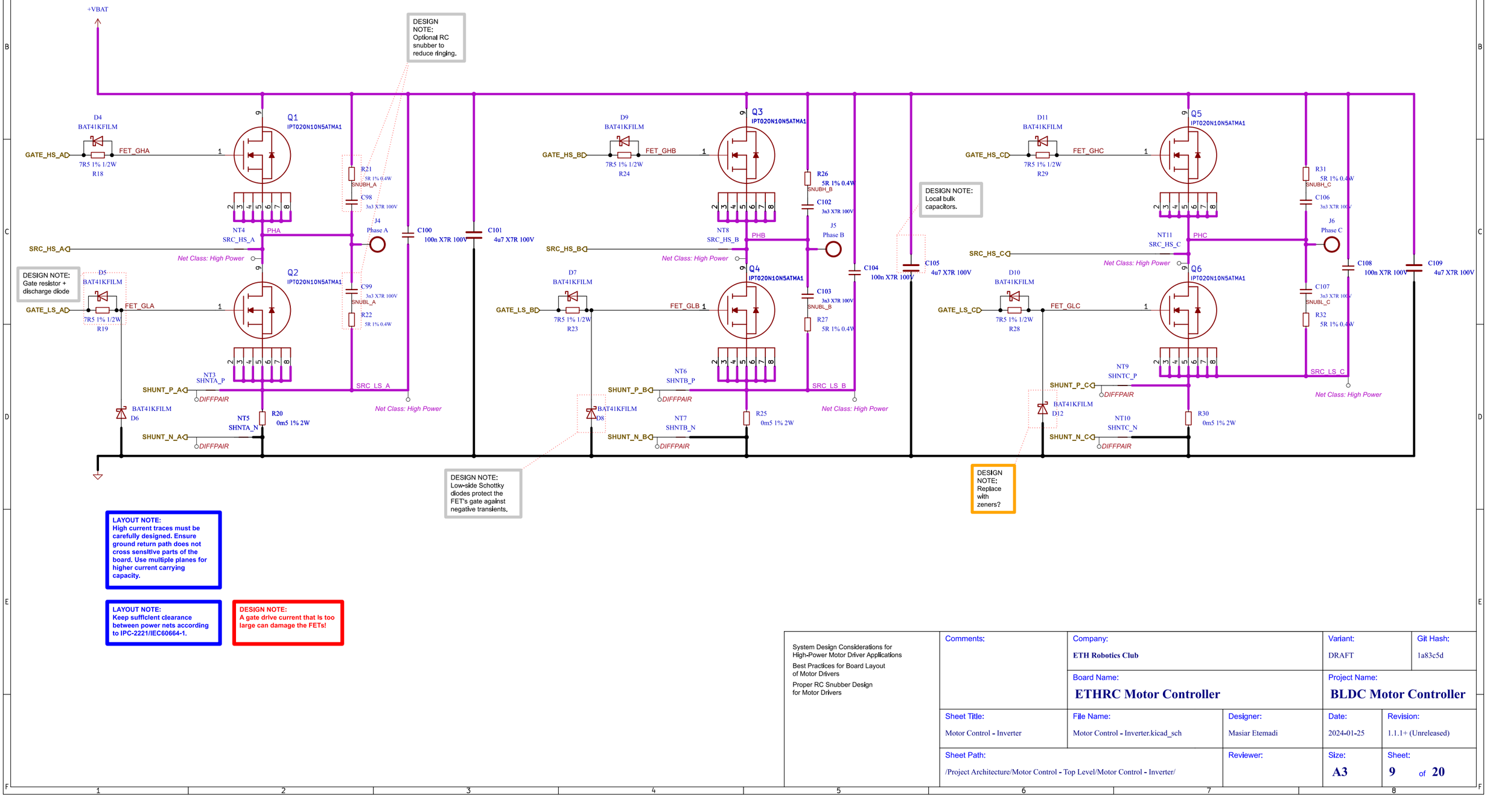
Three-Phase Inverter



Test Points

	Comments:	Company: **ETH Robotics Club**	Variant: DRAFT	Git Hash: 1a83c5d
		Board Name: **ETHRC Motor Controller**	Project Name: **BLDC Motor Controller**	
	Sheet Title: Motor Control - Top Level	File Name: Motor Control - Top Level.kicad_sch	Designer: Masiar Etemadi	Date: 2023-12-20 / Revision: 1.1.1+ (Unreleased)
	Sheet Path: /Project Architecture/Motor Control - Top Level/		Reviewer:	Size: **A3** / Sheet: **8** of **20**

[9] Motor Control - Inverter



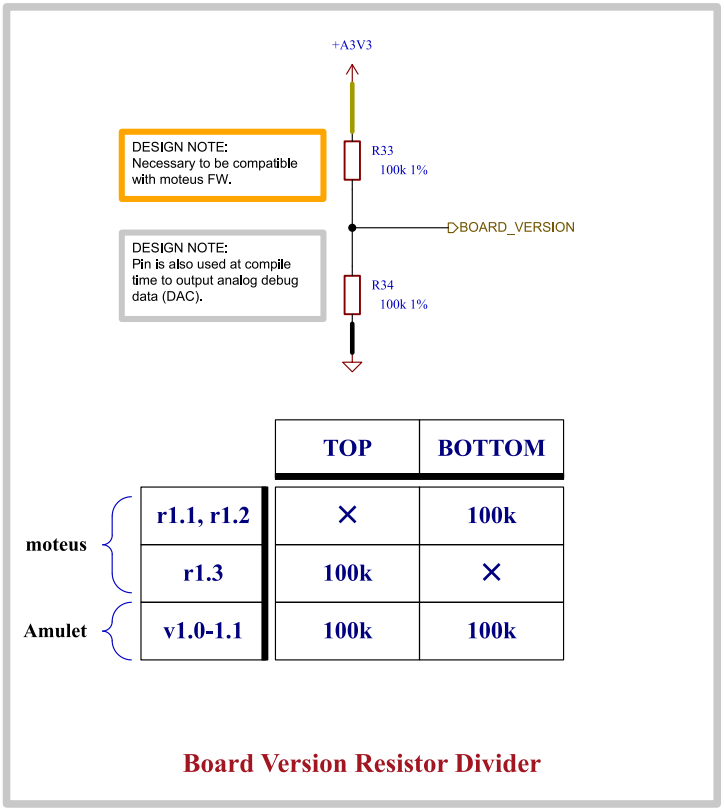
System Design Considerations for High-Power Motor Driver Applications

Best Practices for Board Layout of Motor Drivers

Proper RC Snubber Design for Motor Drivers

Comments:	Company: **ETH Robotics Club**		Variant: DRAFT	Git Hash: 1a83c5d
	Board Name: **ETHRC Motor Controller**		Project Name: **BLDC Motor Controller**	
Sheet Title: Motor Control - Inverter	File Name: Motor Control - Inverter.kicad_sch	Designer: Masiar Etemadi	Date: 2024-01-25	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/Motor Control - Top Level/Motor Control - Inverter/		Reviewer:	Size: **A3**	Sheet: **9** of **20**

[10] Misc - Board Version, DAC

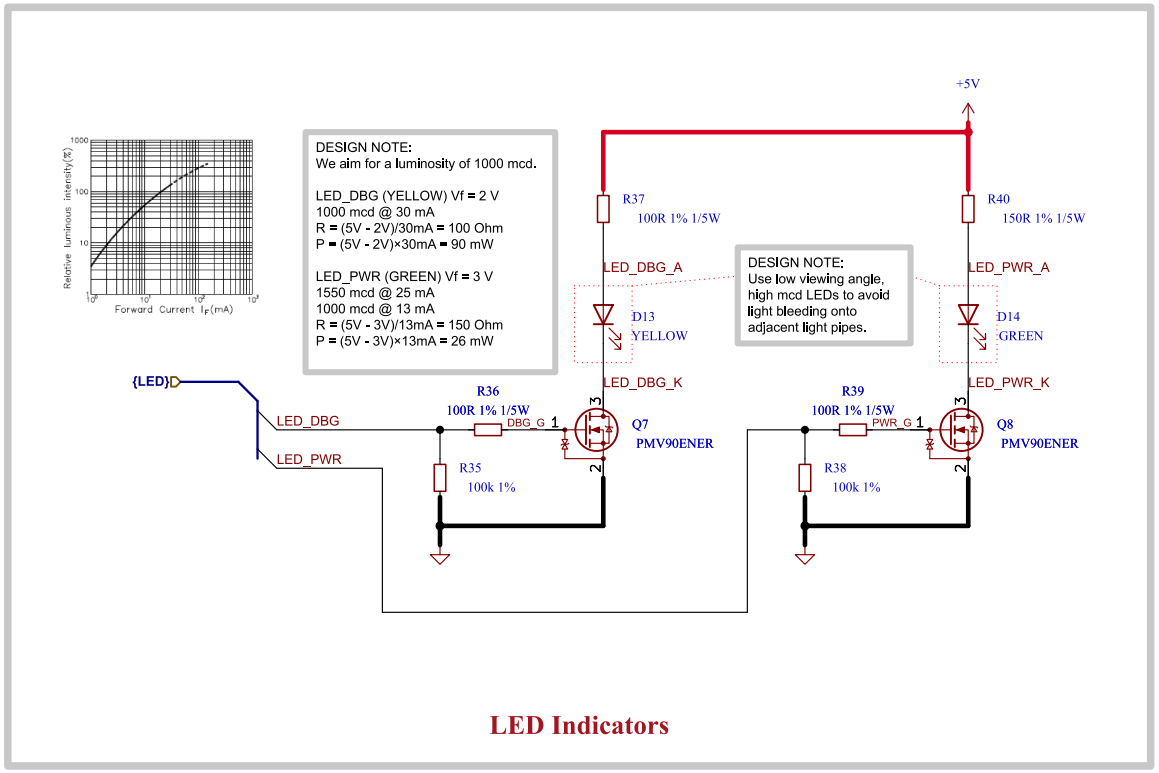


		TOP	BOTTOM
moteus	r1.1, r1.2	×	100k
	r1.3	100k	×
Amulet	v1.0-1.1	100k	100k

Board Version Resistor Divider

Comments:	Company: ETH Robotics Club		Variant: DRAFT	Git Hash: 1a83c5d
	Board Name: ETHRC Motor Controller		Project Name: BLDC Motor Controller	
Sheet Title: Misc - Board Version, DAC	File Name: Misc - Board Version DAC.kicad_sch	Designer: Masiar Etemadi	Date: 2024-04-13	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/Misc - Board Version, DAC/		Reviewer:	Size: A4	Sheet: 10 of 20

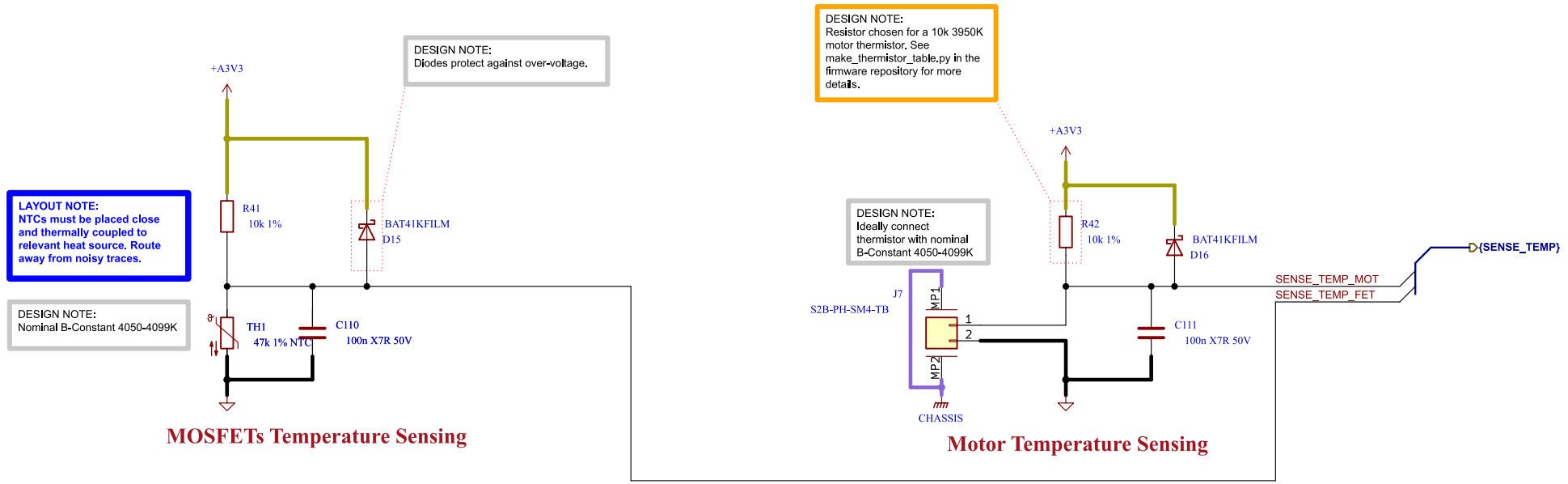
[11] User - LED Indicators



LED Indicators

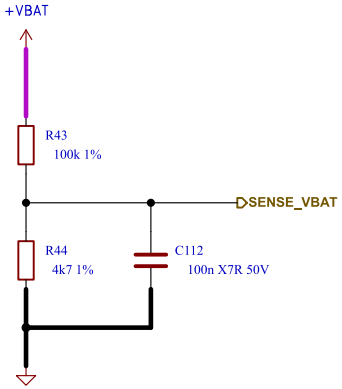
Comments:	Company: **ETH Robotics Club**		Variant: DRAFT	Git Hash: 1a83c5d
	Board Name: **ETHRC Motor Controller**		Project Name: **BLDC Motor Controller**	
Sheet Title: User - LED Indicators	File Name: User - LED Indicators.kicad_sch	Designer: Masiar Etemadi	Date: 2023-12-19	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/User - LED Indicators/		Reviewer:	Size: **A4**	Sheet: **11** of **20**

[12] Sensing - Temperature



Comments:	Company: ETH Robotics Club		Variant: DRAFT	Git Hash: 1a83c5d
	Board Name: ETHRC Motor Controller		Project Name: BLDC Motor Controller	
Sheet Title: Sensing - Temperature	File Name: Sensing - Temperature.kicad_sch	Designer: Masiar Etemadi	Date: 2024-04-13	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/Sensing - Temperature/		Reviewer:	Size: A4	Sheet: 12 of 20

[13] Sensing - Battery



DESIGN NOTE:
1/23 temperature stable voltage divider. Use low ppm resistors.

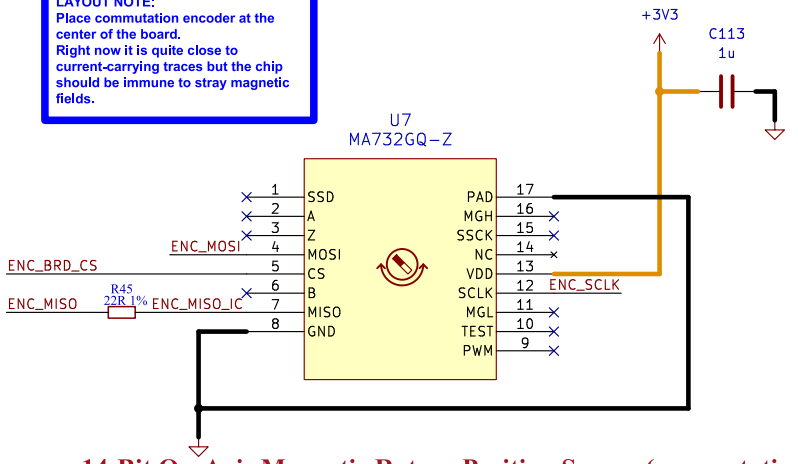
12S: max voltage: 2.19 V
8S: max voltage: 1.95 V
6S: max voltage: 1.46 V

Battery Voltage Sensing

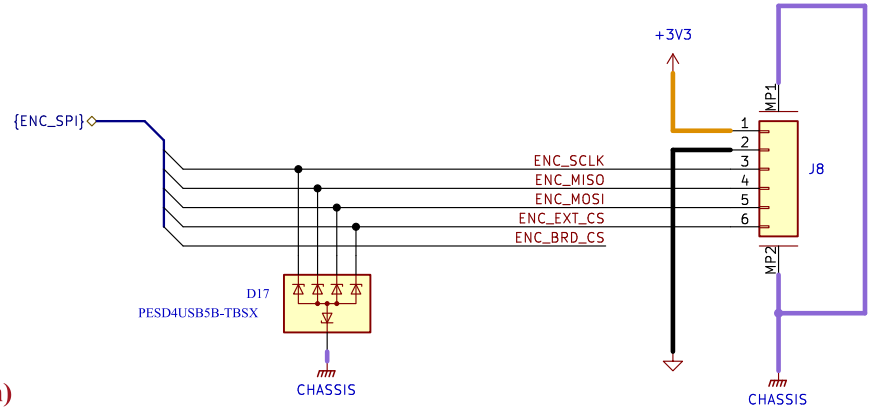
Comments:	Company: ETH Robotics Club		Variant: DRAFT	Git Hash: 1a83c5d
	Board Name: ETHRC Motor Controller		Project Name: BLDC Motor Controller	
Sheet Title: Sensing - Battery	File Name: Sensing - Battery.kicad_sch	Designer: Masiar Etemadi	Date: 2023-10-14	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/Sensing - Battery/		Reviewer:	Size: A4	Sheet: 13 of 20

[14] Sensing - Position

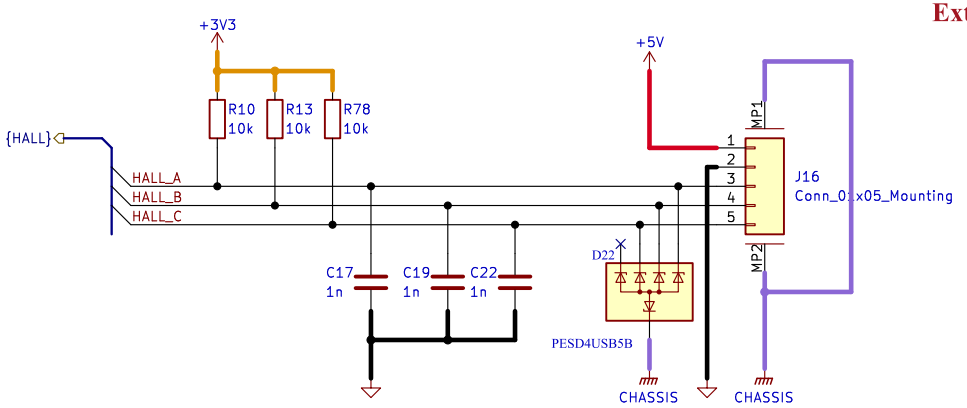
LAYOUT NOTE:
Place commutation encoder at the center of the board.
Right now it is quite close to current-carrying traces but the chip should be immune to stray magnetic fields.



14-Bit On-Axis Magnetic Rotary Position Sensor (commutation)



External 14-Bit On-Axis Magnetic Rotary Position Sensor (disambiguation)



External Hall-Sensor Interface for Commutation

Comments:	Company: **ETH Robotics Club**		Variant: DRAFT	Git Hash: 1a83c5d
	Board Name: **ETHRC Motor Controller**		Project Name: **BLDC Motor Controller**	
Sheet Title: Sensing - Position	File Name: Sensing - Position.kicad_sch	Designer: Masiar Etemadi	Date: 2023-10-14	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/Sensing - Position/		Reviewer:	Size: **A4**	Sheet: **14** of **20**

[15] Comms - EtherCat

ETHERCAT — LAN9253 (SPI PDI, 2 PORTS), MagJacks, SII EEPROM

49.9R terminations to +3V3_PHY (at the LAN9253 pins)

MagJack Connectors (Connector + Magnetics in one)

+3V3_PHY
R59 49.9 1%
R61 49.9 1%
R63 49.9 1%
R65 49.9 1%
ESC_LINK0
R67 330
J9 J0011D21BNL
TD+ 1
TCT 4
TD− 2
RD+ 3
RCT 5
RD− 6
GND 8
GRN 9 10
YLW 11 12
SH1 SHIELD
SH2 SHIELD
C136 10p
C138 10p
C140 10p
C142 10p
C144 0.022u
CHASSIS

+3V3_PHY
+3V3
FB5
600R@100MHz
+1V2_ESC

U9
VDD33 5
VDDIO 14
VDDIO 20
VDDIO 32
VDDIO 37
VDDIO 47
REG_EN 7
VDD33BIAS 58
VDD33TXRX1 51
VDD33TXRX2 64
VDDCR 6
VDDCR 24
VDDCR 38
VDDCR 56
VDD12TX1 59
VDD12TX2
OSCVDD12 3
OSCI 1
OSCO 2
CLK_25/XTAL_MODE crystal mode 8
RST# 11
TESTMODE tie low 41
RBIAS 57
WAIT_ACK/LATCH0 10
D14/TX_SHIFT1 15
D13/TX_SHIFT0 16
D12 21
D11 22
D10 23
A1 25
A3 26
A4 27
OSCVSS 4
VSS(EP) 65
SCK 19
SI/SIO0 17
SO/SIO1 13
SCS# 50
D2/SIO2 12
D3/SIO3 55
IRQ 44
SYNC0/LATCH0 34
SYNC1/LATCH1 18
EESDA/EE_EMUL1 42
EESCL/EE_EMUL2 43
strap E2PSIZE=1 RUNLED/E2PSIZE 45
strap 100FD_B ERRLED/100FD_B 9
strap CHIP_MODE0 LINKACTLED0/CHIP_MODE0 48
strap CHIP_MODE1 LINKACTLED1/CHIP_MODE1 46
TXPA 53
TXNA 52
RXPA 55
RXNA 54
TXPB 62
TXNB 63
RXPB 60
RXNB 61
CS 28
A2/LINKACTLED2 29
WR 30
RD 31
A0/D15 33
D6 36
D7 39
D8 40
D4 49
LAN9253-V/R4X

ESC_SCLK
ESC_MOSI
ESC_MISO
ESC_CS
22R 1% R56
22R at U2
{ESC_SPI}
ESC_IRQ
ESC_SYNC0
ESC_SYNC1
ESC_RST
{ESC_CTRL}
EE_SDA
EE_SCL
ESC_RUN
ESC_ERR
ESC_LINK0
ESC_LINK1
ECAT_P0_TX+
ECAT_P0_TX−
ECAT_P0_RX+
ECAT_P0_RX−
ECAT_P1_TX+
ECAT_P1_TX−
ECAT_P1_RX+
ECAT_P1_RX−

25 MHz crystal
Y1 25MHz 8pF
C116 10p C0G
C117 10p C0G
C119 1u
C120 100n
ESC_OSCVDD12
ESC_XI
ESC_XO
ESC_CLK25
ESC_RST
ESC_RBIAS
unused in SPI mode
R54 1k
R55 12.1k 1%
R57 10k
R58 10k
+3V3
R51 10k
C118 100n
Mode strap, reset, SPI MISO series

DESIGN NOTE:
WP low: master can write the SII

SII EEPROM (I2C address 0x50)
+3V3
100n C115
R46 10k
U8
VCC 4
WP 5
VSS 2
SCL 1
SDA 3
24LC32AT-I/OT
R50 2.2k
R52 2.2k
EE_SCL
EE_SDA

+3V3
C121 1u
C123 100n
C125 100n
C127 100n
C129 100n
C131 100n
C132 100n
+3V3: VDD33 (5) 1u+100n, VDDIO (14, 20, 32, 37, 47) 100n each, U3 100n

+3V3_PHY
C133 100n
C134 100n
C135 100n
+3V3_PHY: pins 51, 58, 64, T1/T2 centre taps, FB5 output

+1V2_ESC
C122 100n
C124 100n
C126 100n
C128 100n
C130 100n
C161 1u
+1V2_ESC (internal regulator): C161 1u low-ESR (DS00003421B 4.1.1) at VDDCR pin 6; 24, 38, 56, 59 100n each

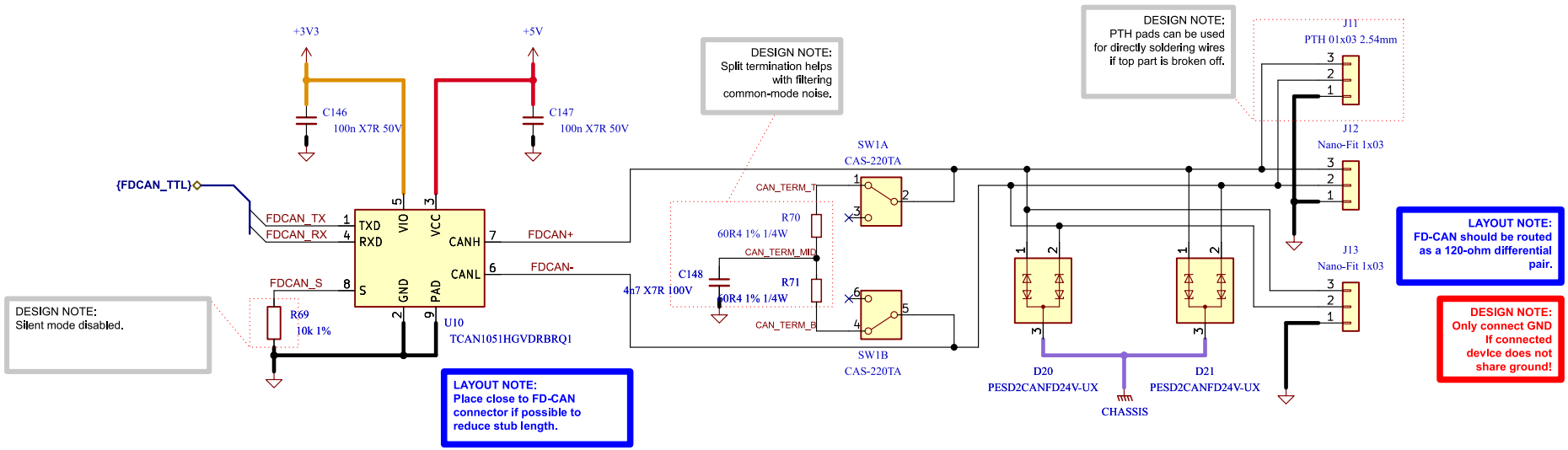
EtherCAT indicators + strap resistors (straps read at reset)
ESC_ERR
R47 2.2k
R48 10k
D18 Red
+3V3
R49 10k
D19 Green
R53 2.2k
ESC_RUN

+3V3_PHY
R60 49.9 1%
R62 49.9 1%
R64 49.9 1%
R66 49.9 1%
ESC_LINK1
R68 330
J10 J0011D21BNL
C137 10p
C139 10p
C141 10p
C143 10p
C145 0.022u
CHASSIS

LAYOUT NOTE: ETHERCAT
* TX/RX pairs 100 ohm differential, matched in-pair, short; 49.9R at the U2 pins.
* Keep PHY side and cable side of T1/T2 apart; no GND plane under the cable side and the GH connectors except the Bob Smith cap return; C94/C97 right at the centre taps.
* Y1 + caps and R85 (RBIAS) tight to their pins; exposed pad via array to GND.

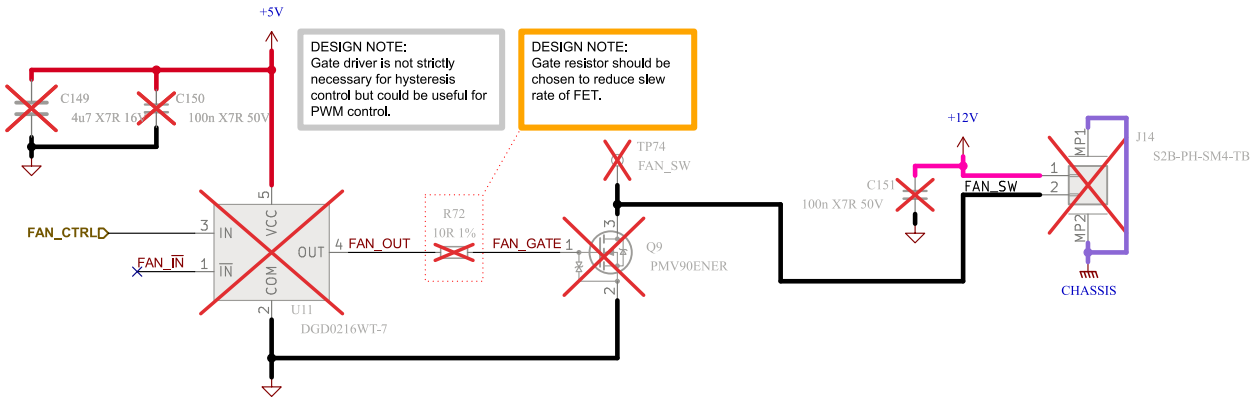
Comments:	Company: ETH Robotics Club		Variant: DRAFT	Git Hash: 1a83c5d
	Board Name: ETHRC Motor Controller		Project Name: BLDC Motor Controller	
Sheet Title: Comms - EtherCat	File Name: Comms - EtherCat.kicad_sch	Designer: Masiar Etemadi	Date: 2025-01-12	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/Comms - EtherCat/		Reviewer:	Size: A3	Sheet: 15 of 20

[16] Interface - FD-CAN



Comments:	Company: ETH Robotics Club		Variant: DRAFT	Git Hash: 1a83c5d
	Board Name: ETHRC Motor Controller		Project Name: BLDC Motor Controller	
Sheet Title: Interface - FD-CAN	File Name: Comms - FD-CAN.kicad_sch	Designer: Masiar Etemadi	Date: 2023-10-15	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/Comms - FD-CAN/		Reviewer:	Size: A4	Sheet: 16 of 20

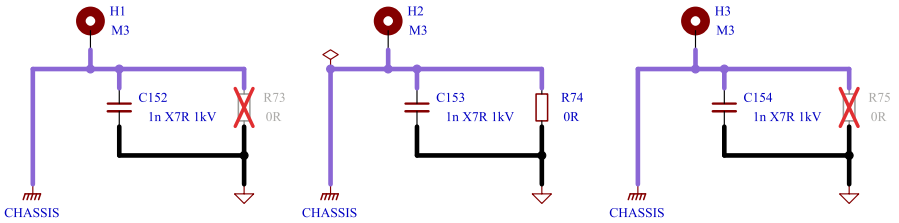
[17] Interface - Fan Control



Low-side Fan Switch and Gate Driver

Comments:	Company: **ETH Robotics Club**		Variant: DRAFT	Git Hash: 1a83c5d
	Board Name: **ETHRC Motor Controller**		Project Name: **BLDC Motor Controller**	
Sheet Title: Interface - Fan Control	File Name: Fan Control.kicad_sch	Designer: Masiar Etemadi	Date: 2023-11-19	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/Fan Control/		Reviewer:	Size: **A4**	Sheet: **17** of **20**

[19] Misc - Holes, Fiducials



DESIGN NOTE:
Optional RC filter for chassis grounding. Capacitors offer multi-point low-impedance connections at high-frequencies.

DESIGN NOTE:
0-Ohm resistor provides low-impedance bond between chassis and signal GND. Be aware of ground loops through cable shield.

Mounting Holes and Chassis Connection

Front

FM1 FM3 FM5

Back

FM2 FM4 FM6

Fiducial Markers

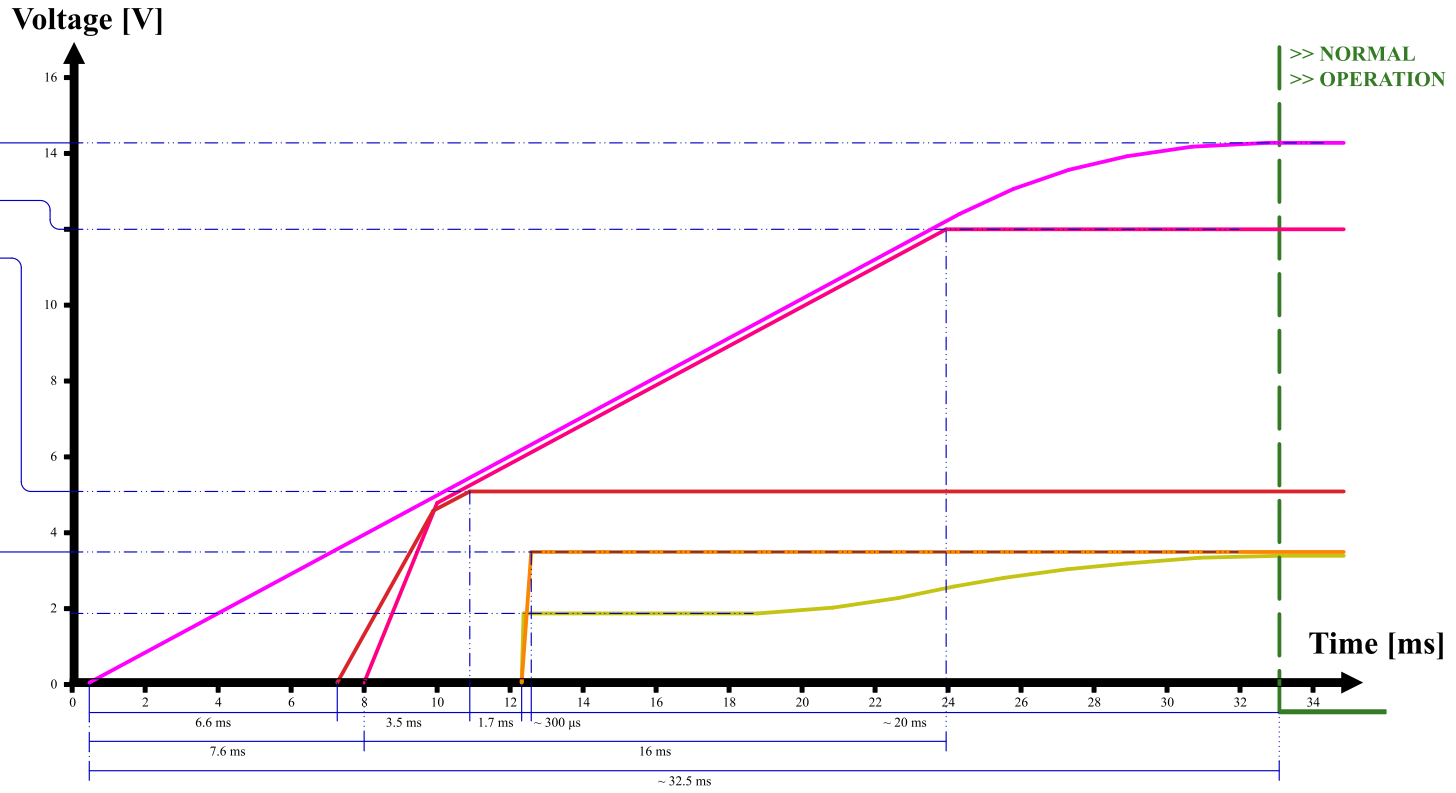
Comments:	Company: ETH Robotics Club		Variant: DRAFT	Git Hash: 1a83c5d
	Board Name: ETHRC Motor Controller		Project Name: BLDC Motor Controller	
Sheet Title: Misc - Holes, Fiducials	File Name: Misc - Holes Fiducials.kicad_sch	Designer: Masiar Etemadi	Date: 2023-10-22	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Project Architecture/Misc - Holes, Fiducials/		Reviewer:	Size: A4	Sheet: 19 of 20

[20] Power - Sequencing

NAME	SOURCE	LEVEL
+VBAT	BATTERY	12 - 44V
+12V	LMR36006	12V ± 1.5%
+5V	LMR36506	5V ± 1.5%
+3V3	TPS62172	3.3V ± 3%
+A3V3	LP2992	3.3V ± 0.5%

DESIGN NOTE:
Power sequencing graph measured experimentally.

DESIGN NOTE:
Start-up time of 12V rail depends on +VBAT charging time, which itself depends on the precharge circuit.



Comments:	Company: ETH Robotics Club		Variant: DRAFT	Git Hash: 1a83c5d
	Board Name: ETHRC Motor Controller		Project Name: BLDC Motor Controller	
Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Masiar Etemadi	Date: 2024-03-12	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 20 of 20

[21] Revision History

Version 1.0.0 - 2023-12-12

Added

- TVS protection and termination switch to FD-CAN.
- Low-side switched 12V 600 mA source for external fan.
- LDO for analog supply.
- One TVS diode per power connector.
- Second on-board I2C magnetic encoder for disambiguation.
- ESD protection to all interfaces.
- Over-voltage protection on thermistor ADC inputs.
- Pi filters to inputs of buck regulators and MCU analog supply.
- Decoupling caps next to power pins of connectors.

Changed

- CPH-CPL capacitor to 47 nF (gate driver).
- FD-CAN transceiver IC.
- FETs for top cooled variant.
- Input power TVS diode to bidirectional.
- Moved SOx low-pass filter to MCU section.
- PWM_PHASEA with PWM_PHASEC on STM32G474 pinout for easier routing.
- RS422 pinout on connector.
- Buck regulators to optimize for low noise.

Version 1.0.1 - 2024-01-25

Added

- Controller target specifications.
- Credits to moteus on cover page.
- Optional RC-Snubber to power stage.

Fixed

- Chassis guard ring to go around the board.
- CAN and power TVS diodes now go to chassis.
- Clearance between nets to respect IEC60664-1 where possible.
- Comment on precharge.
- Power TVS diode reference designator from "U" to "D".

Changed

- 5V 300 mA buck converter with 600 mA version.
- Chassis-GND capacitor by 1nF 1kV.

Version 1.0.2 - 2024-03-12

Changed

- Power sequencing graph according to experimental data.

Version 1.1.0 - 2024-04-13

Added

- RC snubber passive values.

Fixed

- More vias for VBUS and LMR36006 GND pads.
- Board version voltage reference from +3V3 to +A3V3.

Changed

- Motor thermistor resistor divider to 2kOhm for a 10k 3950K thermistor.

Version 1.1.1 - 2025-01-18

Fixed

- Replace non-stocked components.

Comments:	Company: ETH Robotics Club		Variant: DRAFT	Git Hash: 1a83c5d
	Board Name: ETHRC Motor Controller		Project Name: BLDC Motor Controller	
Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Masiar Etemadi	Date: 2024-01-03	Revision: 1.1.1+ (Unreleased)
Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 21 of 20